

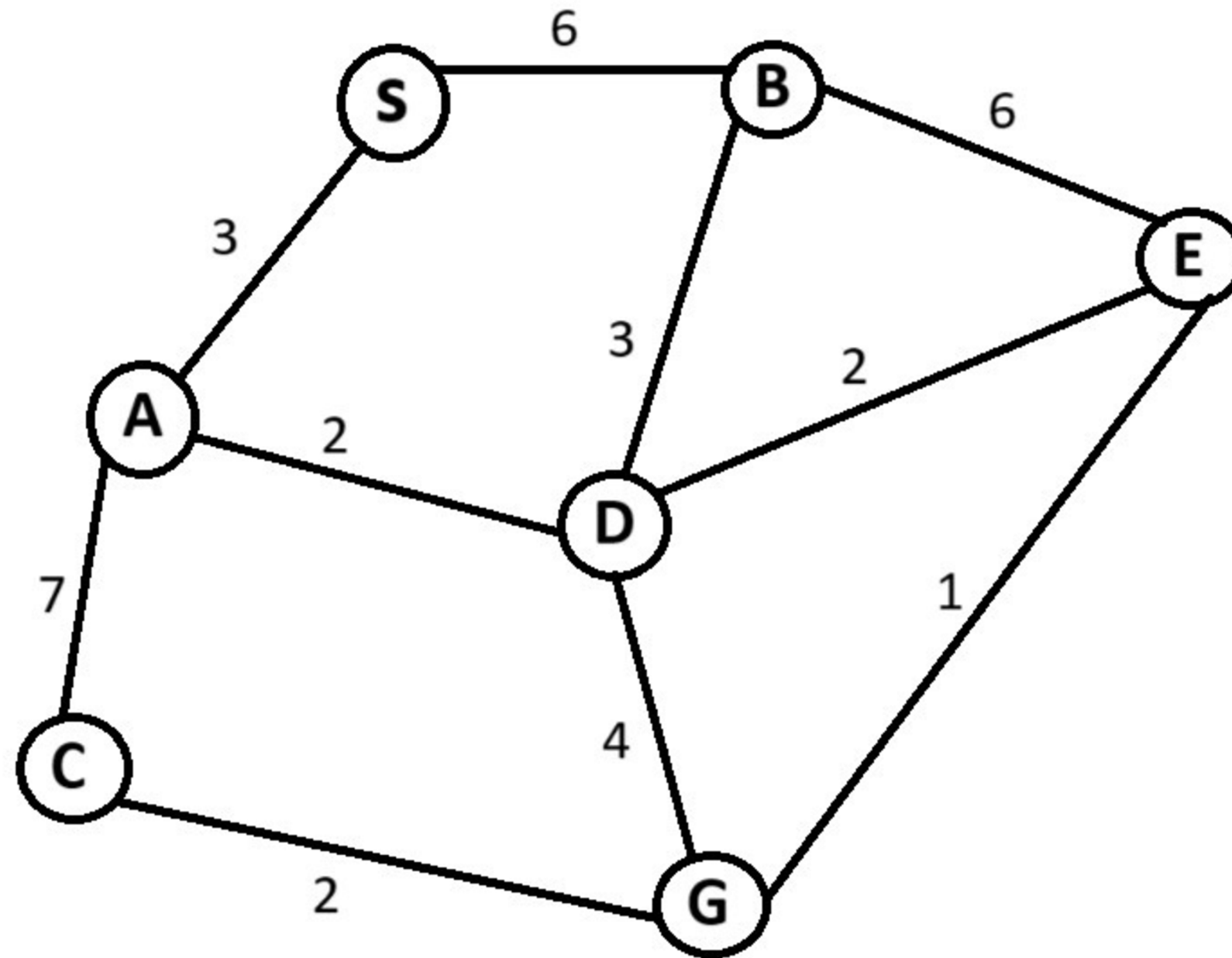
BRANCH & BOUND ALGORITHM

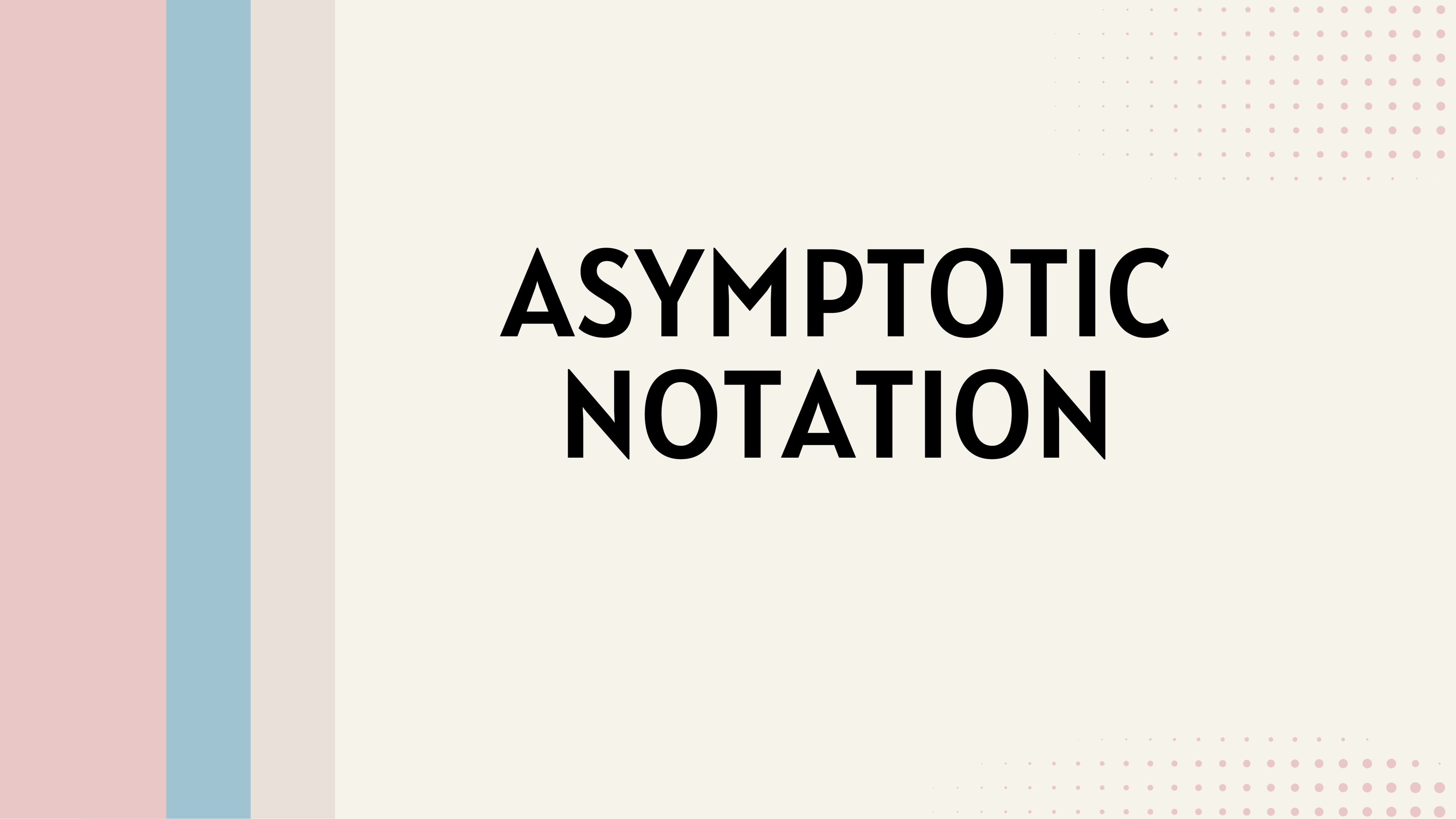
BRANCH & BOUND ALGORITHM

- Branching is the process of generating subproblems.
- Bounding refers to ignoring partial solutions that cannot be better than the current best solution.
- It is a search procedure to find the optimal solution.
- It eliminates those parts of a search space which does not contain better solution(Pruning).

In this method we basically extend the cheapest partial path.

BRANCH & BOUND ALGORITHM



The background features three vertical bars on the left: a wide pink one, a narrower blue one, and a thin light beige one. In the top right and bottom right corners, there are rectangular areas filled with a grid of small, light pink dots.

ASYMPTOTIC NOTATION

ASYMPTOTIC NOTATION

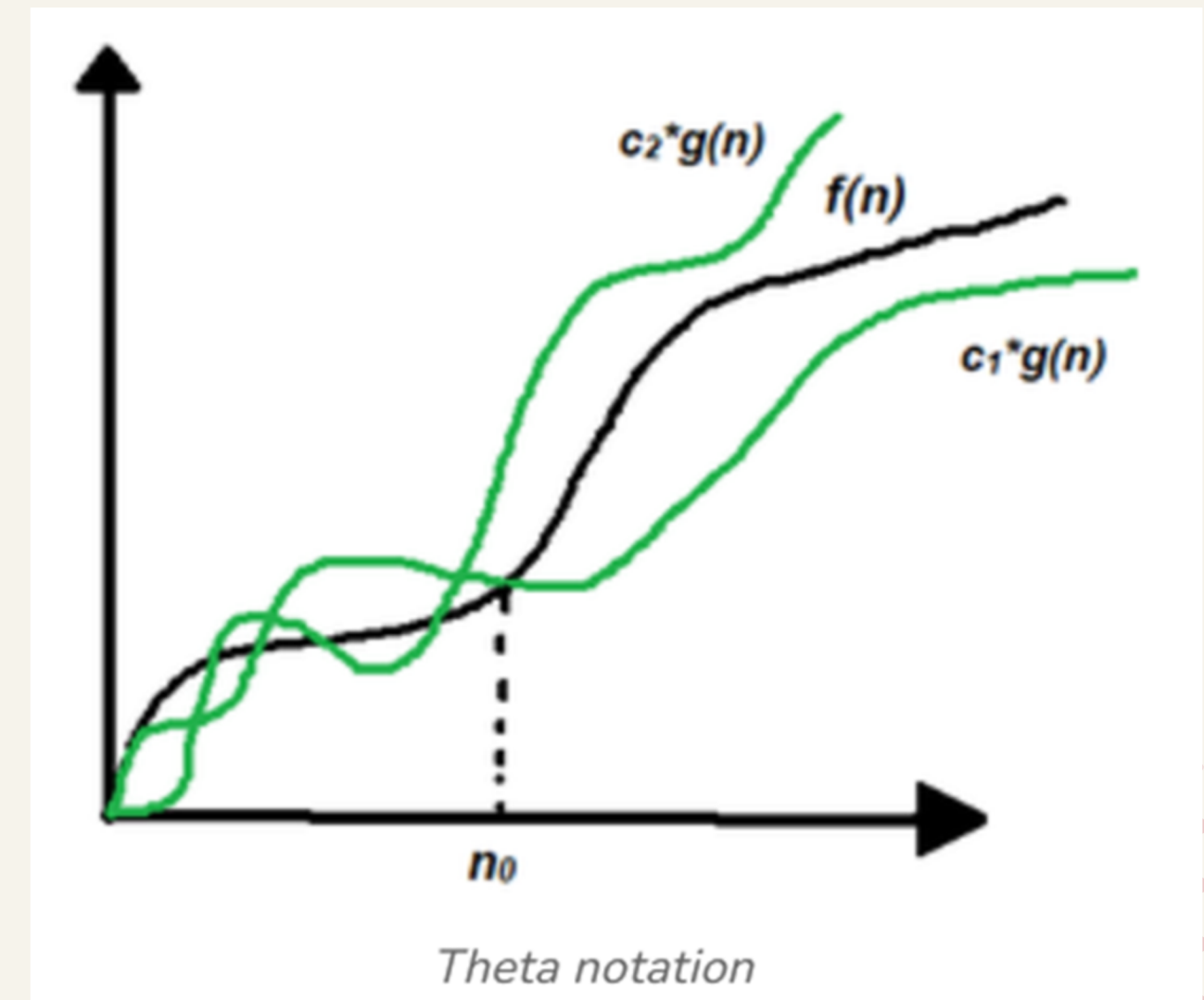
Asymptotic notations are mathematical tools to represent the time complexity of algorithms for asymptotic analysis.

There are mainly three asymptotic notations:

1. Theta Notation (Θ -notation)
2. Big-O Notation (O -notation)
3. Omega Notation (Ω -notation)

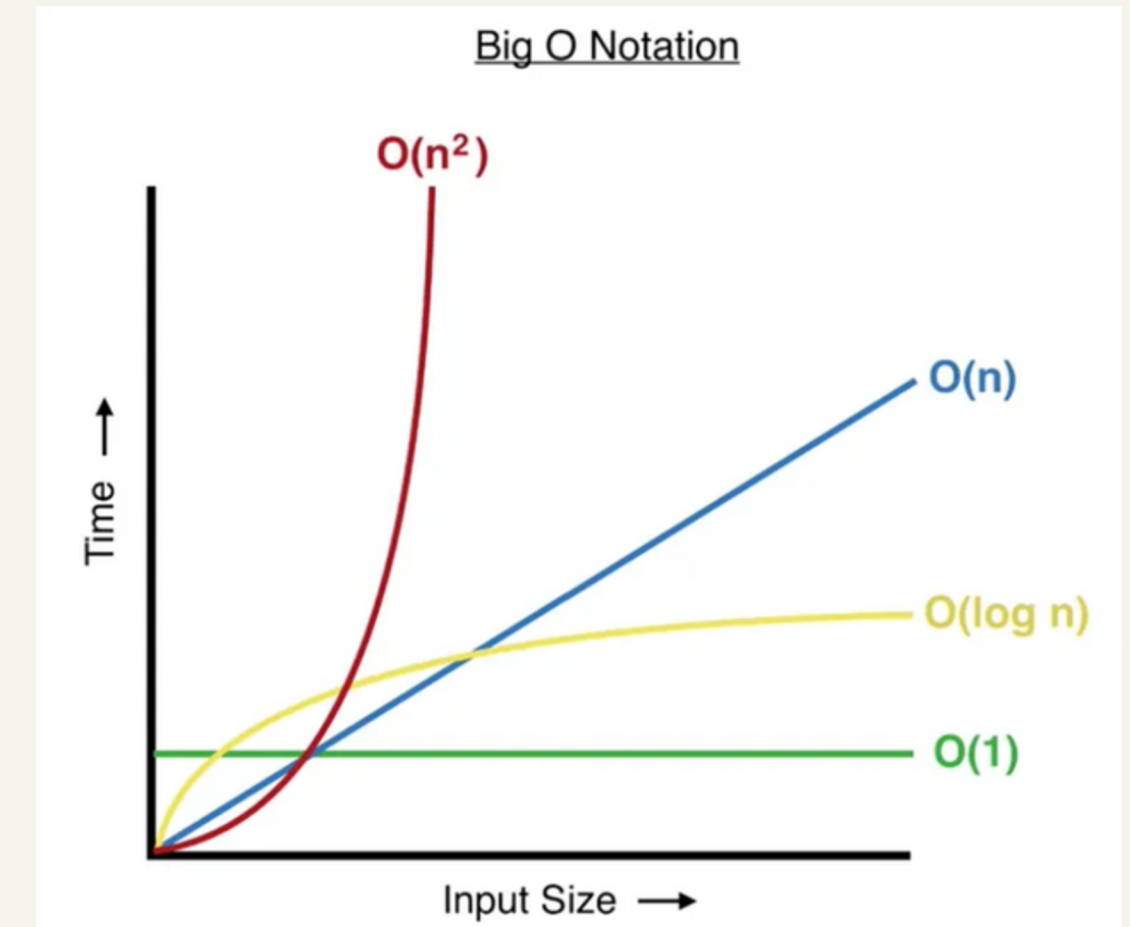
I. Theta Notation (Θ -Notation):

it represents the upper and the lower bound of the running time of an algorithm, it is used for analyzing the average-case complexity of an algorithm.

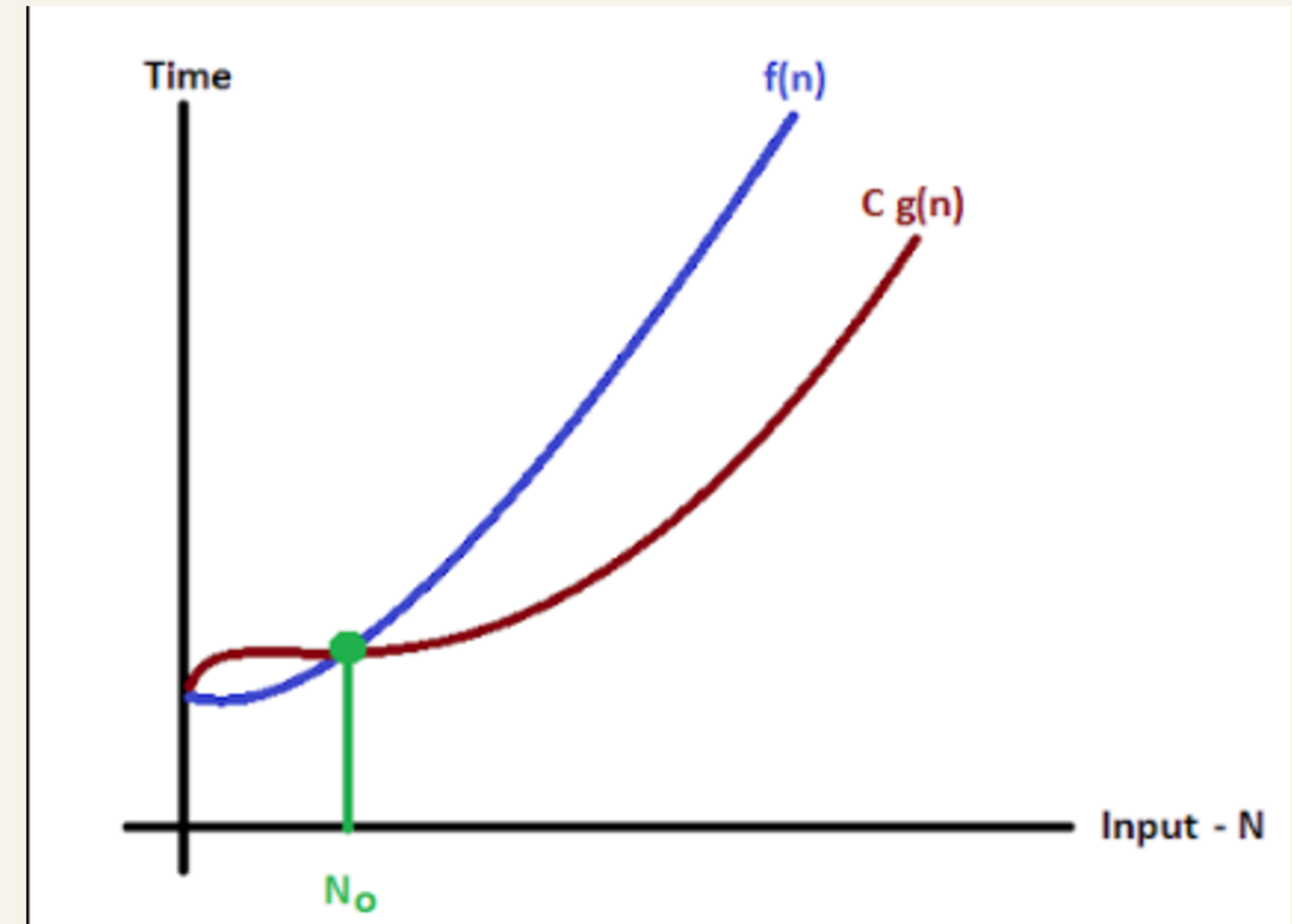


2. Big-O Notation (O-notation):

Big-O notation represents the upper bound of the running time of an algorithm. Therefore, it gives the worst-case complexity of an algorithm.



3. Omega Notation (Ω -Notation):
Omega notation represents the lower bound of the running time of an algorithm. Thus, it provides the best case complexity of an algorithm.



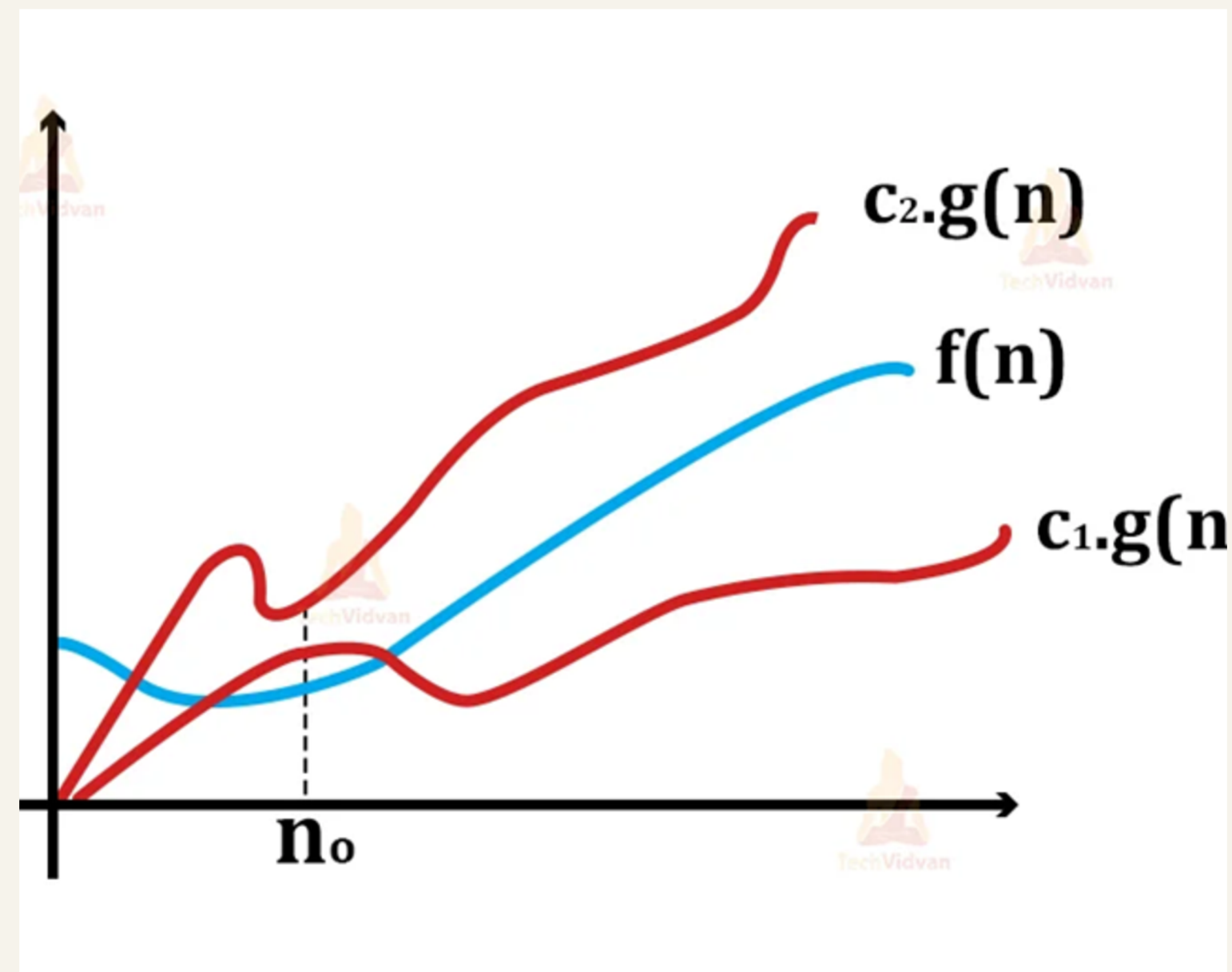
GROWTH RATE

GROWTH RATE

Rate of growth is defined as the rate at which the running time of the algorithm is increased when the input size is increased.

The growth rate could be categorized into two types: linear and exponential. If the algorithm is increased in a linear way with an increasing in input size, it is linear growth rate.

GROWTH RATE





BINARY SEARCH

I. Introduction to Binary Search

Definition and working principle

Comparison with linear search

Advantages and limitations



2. Applications of Binary Search

Searching in sorted arrays

3. Binary Search Algorithm

Step-by-step explanation with pseudocode

Dry-run examples on different datasets

Time complexity analysis

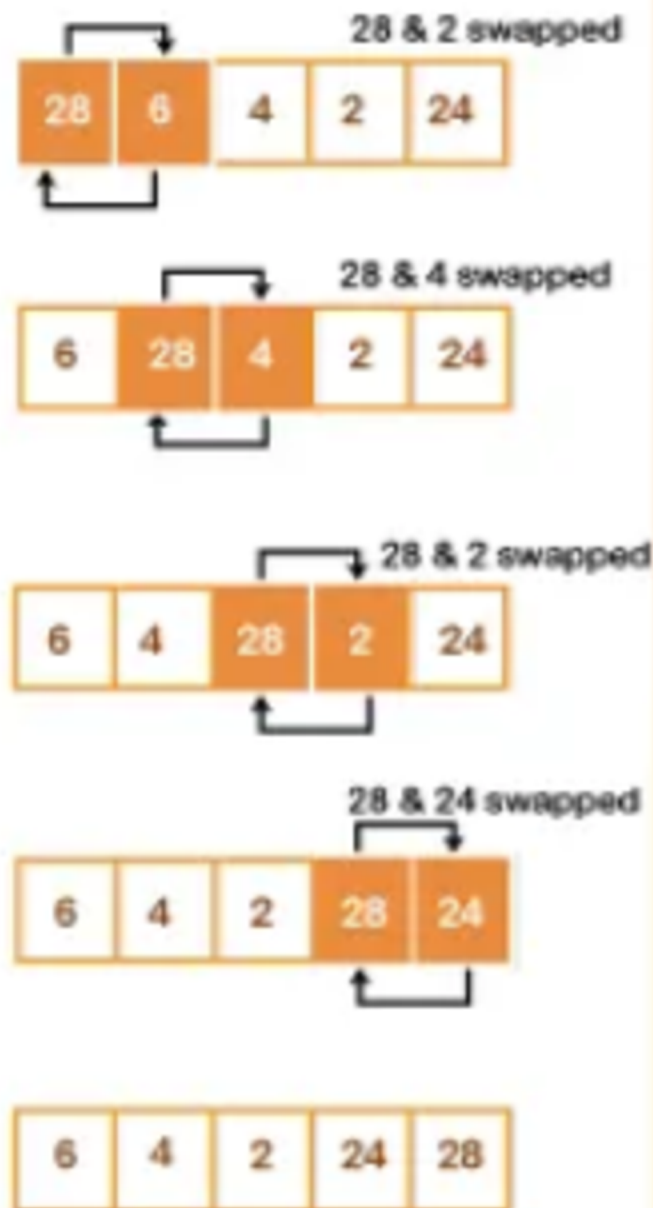
BUBBLE SORT

Bubble Sort is a simple comparison-based sorting algorithm that repeatedly steps through a list, compares adjacent elements, and swaps them if they are in the wrong order. The process is repeated until no more swaps are needed, meaning the list is sorted.

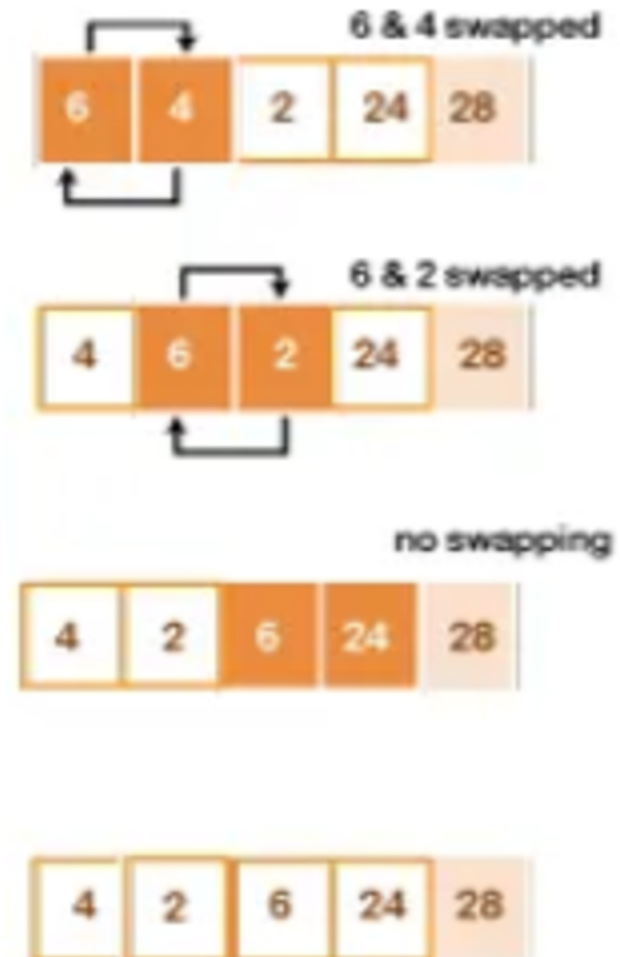
Bubble Sort in Java

28 6 4 2 24

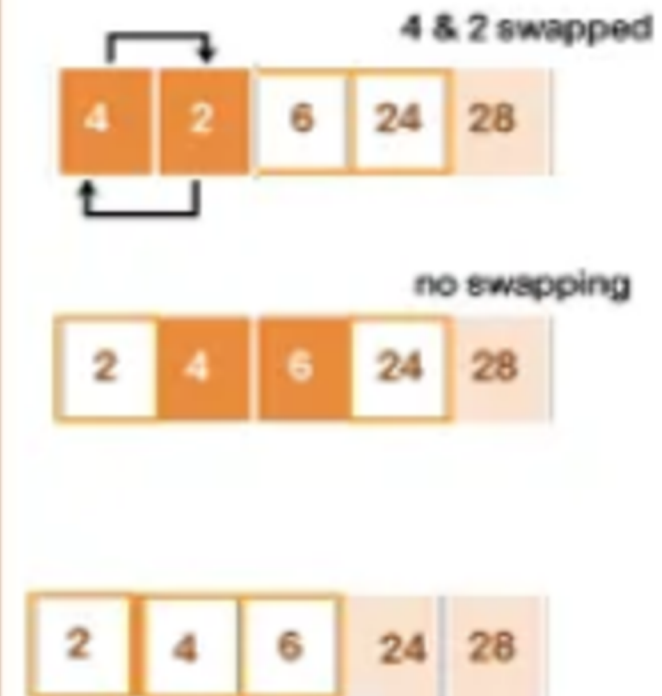
Pass 1



Pass 2



Pass 3



Pass 4



MERGE SORTING

14

Merge sort is a sorting algorithm that uses the divide-and-conquer technique to organize a list of items in order. It's considered one of the most efficient sorting algorithms.

How does merge sort work?

1. Divide

The array is broken down into smaller and smaller subarrays until each subarray has only one item.

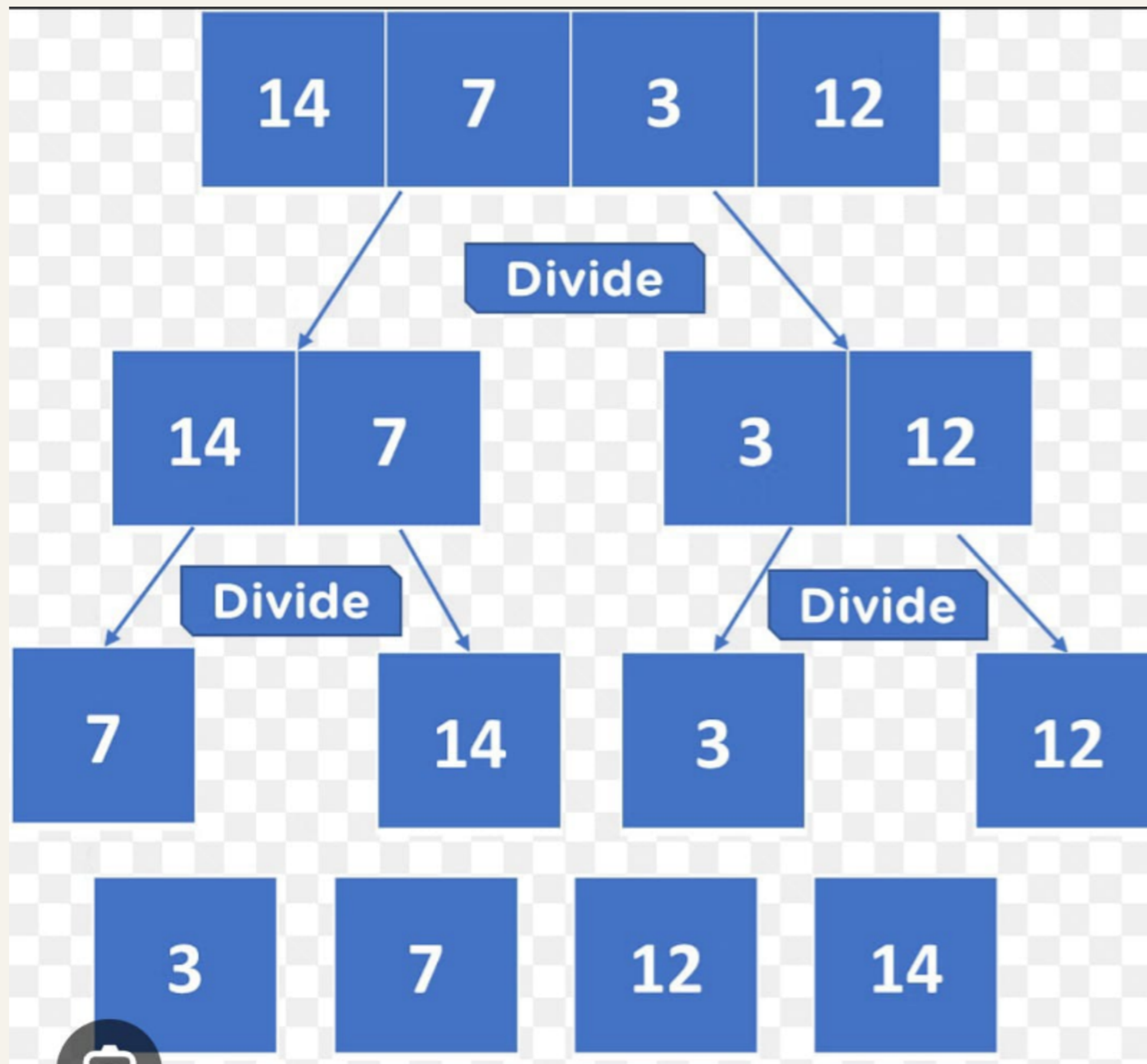
2. Conquer

The subarrays are merged back together by putting the lowest values first. This process continues until all elements are combined into a single sorted list.

When is merge sort useful?

Merge sort is efficient for sorting large data sets.

It's popular in languages like Lisp, where sequentially accessed data structures are common.



The background features three vertical bars on the left: a wide pink bar, a medium blue bar, and a narrow beige bar. On the right side, there are two rectangular areas filled with a grid of small pink dots, one in the top right and one in the bottom right.

THANK YOU